From Diagnosis to Fixes: Mechanisms That Close, and Widen, the Gaps of Language-Model Agent Teams

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Abstract

Teams of language-model agents are known to fail in predictable ways: they do not pool private information, they trust bad information, and, when trained with the wrong incentives, they stop cooperating. We move from diagnosing these failures to testing fixes for them. In controlled, pre-registered experiments with Claude Haiku 5.5 teams, we measure each fix as the share of a gap closed against an explicit ceiling (the information-pooled team, the team without a liar, a single agent with all facts), and pair each mechanism with a prompt-only control. Three results stand out. First, when messages cost time, adding communication can lower team output: a request board widened the pooling gap (gap closed -0.18 at four agents, and about the same at 8, 16 and 32 agents) and an instruction to share first nearly doubled sharing yet lowered output. Second, a public track record of each sender's past accuracy, computed by the harness, recovered 0.32 $[0.20, 0.44]$ of the damage done by a lying agent at no cost when no liar was present, and kept recovering about a quarter to a third of it in teams of up to 32 agents, whereas a private, costly verification action caught most lies and recovered nothing; a 2×2 design shows that being public and being free each contribute about equally. Third, in our hidden-profile vote a structured table of the facts posted so far raised the win rate from 0.88 to 0.97 without increasing disclosure; this did not replicate on HiddenBench's 65 tasks, where only an instruction to speak up helped, by raising disclosure. Training a 3B open model with reinforcement learning on the same cooperation task shows the incentive side: individual reward found a guessing loophole and abandoned sharing; pricing the loophole, with the same individual reward, kept the agents sharing; and with credit given, RL beat an imitation-learned starting point ($+0.068$ $[0.026, 0.112]$ welfare per agent-turn pooled over seeds; best runs $+44\%$), with as much variation between training seeds as between rewards. We release the environments, analysis plans and code that regenerates every reported number.

1 Introduction

Multi-agent deployments of language models are growing, from orchestrators with parallel workers [2] to large groups solving open-ended problems. Their failures are now well documented. Groups of agents fail hidden-profile tasks, in which the right answer needs facts that only individual members hold: across 15 models, groups reach about 30% accuracy where a single agent with all facts reaches 81% [13], and Anthropic's red team reports 17–36% for most models [3]. Agents conform [28], adopt a single false testimony more readily than true ones [29], and under reinforcement learning become opportunistic in social dilemmas [9, 21]. Studying such failures empirically is part of the science of machine behaviour [23]. The red team names the fixes as open problems: mechanism design, reputation, and environments that shape cooperation [3].

This paper tests fixes. For each failure we build a small, fully specified environment, freeze an analysis plan before any run, and compare a baseline team with teams given one mechanism at a time. We report every fix as the *share of a gap closed* between the baseline and an explicit ceiling, paired on seed or episode, and pair each mechanism (an *affordance*: a new action, an observation line, a harness rule) with a prompt-only *instruction* control. Our contributions:

1. **Communication has a price.** When a message costs a turn, a request board and an instruction to share first both lower team output, while a plain shared board is best (§4.1).

2. **Public trust beats private verification.** A harness-computed public track record recovers a fifth to a third of liars' damage at no cost, at every team size up to 32; private, costly checks catch lies but do not help; publicity and cost contribute about equally (§4.2, §4.6).

3. **A fix that did not replicate.** A structured fact table improved our hidden-profile vote without increasing disclosure, but not HiddenBench's tasks, where an instruction to speak up helped by raising disclosure (§4.3, §4.4).

4. **Incentives decide whether trained agents cooperate, and a mechanism fix suffices.** With RL on the same task, individual reward produces selfish trial and error through a guessing loophole; pricing the loophole keeps agents cooperative under the same reward; with credit given, RL beats imitation (and nearly does with the loophole priced), though single training seeds can hide the gain (§5).

2 Related work

Information pooling in agent teams. The hidden-profile paradigm [24] has been ported to LLM groups [3, 13]. Diagnoses stress early convergence on shared evidence [13]; teams average expert and non-expert views instead of weighting expertise [18]; up to 72% of issue-critical facts can be lost in deliberation [25]. An LLM facilitator raised how much information human groups shared without improving their decisions [1], the mirror image of our vote result. Adaptive communication control can beat full information on hidden-profile tasks [4], and structured deliberation protocols can beat unstructured debate at a large token cost [22].

Communication cost and structure. Pruning redundant messages and agents saves tokens at comparable accuracy [26, 32]; across 260 configurations, multi-agent architectures range from large gains to large losses relative to a single agent depending on the task [12]. These measure cost in tokens; in our tasks a message costs task time.

Deception and trust. Defences learn credibility scores from past contributions [6], detect threats and rank neighbours [7], or rectify misleading content [16]. We compare a public, harness-computed signal with private verification at matched cost, a contrast we did not find studied. Honest-signalling theory predicts that verifiability matters [31]; our results suggest that who sees the verification matters too.

Training agent teams. Multi-agent RL for LLM collaboration includes MAPoRL [19], MAGRPO [15] and AT-GRPO [33]; counterfactual credit addresses free-riding [8, 14]. In social dilemmas, RL-trained LLM agents become opportunistic [21] and most societies fail to sustain a commons [20]. Grading that never penalises wrong answers rewards guessing [11]; we observe the multi-agent analogue, a producer-scrounger outcome [5].

3 Setup

The clue board. N agents share a board of $2N + 2$ open subtasks, each worth 1, 2 or 3 points and open for 30 turns. A subtask hides an integer $0 \le x < 30$ fixed by two clues of the form "$x \bmod p = r$" (primes from 3 to 13); the clues are dealt to different agents, so no agent can solve a subtask alone, and with one clue known 2 to 10 candidates remain (most often 2, 3 or 6). Each turn an agent opens a subtask, submits an answer (a wrong answer costs 0.5), shares a clue, or waits. Shared clues go to a channel; here a *blackboard* every agent reads. Welfare is points per agent-turn; the *pooled share* divides it by the welfare the team would earn if every clue were known to all, so 1 is the information ceiling. A scripted liar, when present, posts a false residue for a subtask it is not working on every other turn.

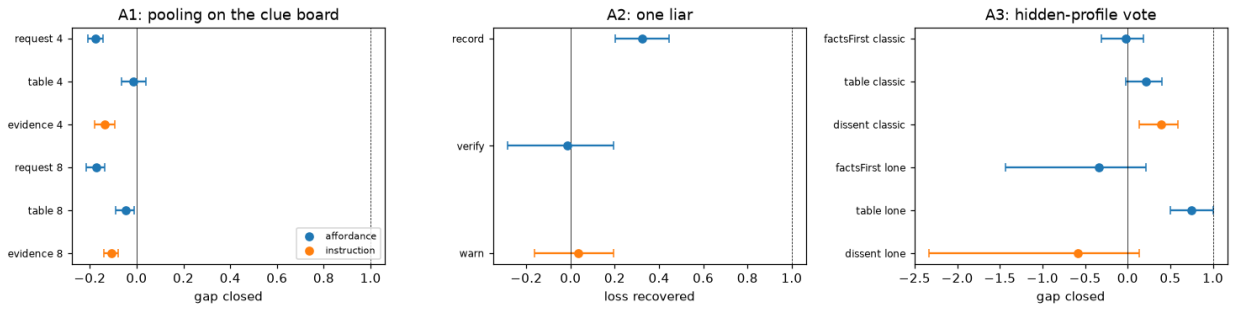


Figure 1: Share of the gap closed (A1, A3) or of the liar's damage recovered (A2) per arm, with 95% intervals. Blue: affordances; orange: prompt-only instructions. 0 is the baseline, 1 the ceiling.

The hidden-profile vote. Four agents choose between two options in one of three domains (hiring, investment, property). Facts held by everyone favour option B (B +3; A +1 and −1); each agent also holds two private facts favouring A, so the full totals are A +8, B +3, while each agent alone sees A +2, B +3. Agents post one fact (published verbatim) or a short opinion per round for three rounds, then vote privately; success is a strict majority for A. In the *lone dissenter* variant one agent also holds a fact that rules B out. The ceiling is one agent given every fact.

Agents and protocol. Claude Haiku 5.5 with low effort and structured JSON replies; clue-board agents see their full history, vote agents a short note plus the whole board. Replies are cached, so every rerun is byte-identical. Each analysis plan (arms, seeds, contrasts, multiple-comparison families) was hashed and committed before its runs; the board runs used 10 seeds (900–909) per arm and team size, the vote 100 episodes per arm and variant on shared episode seeds. Gap closed is the mean paired difference (arm minus baseline) divided by (ceiling minus baseline mean); loss recovered is 1− (arm's mean honest loss / baseline's), where honest loss is the honest agents' welfare without the liar minus with it. Intervals are 95% bootstrap over seeds or episodes; primary tests are Holm-corrected within each part [17]. All findings are exploratory.

4 Fixes for Claude teams

Figure 1 summarises the three parts.

4.1 Pooling on the clue board: communication has a price

The baseline team earned a pooled share of 0.45 at four agents. A *request board* let an agent post that it needed clues for its open subtask, with holders shown how many of the missing clues they had. Agents used it heavily (42 asks per board, about a quarter of their turns), sharing did not rise, and the gap widened: −0.18 $[-0.21, -0.14]$ at four agents and −0.17 $[-0.21, -0.14]$ at eight (Holm $p = 0.008$ for both). An instruction to post every held clue first nearly doubled sharing (0.40 to 0.73 shares per held clue) and also lowered output (-0.14 $[-0.18, -0.09]$). A deduplicated table of the team's known clues had no effect at four agents (-0.01 $[-0.06, 0.04]$). Sharing costs a turn on this board, and the extra communication spent turns that solving needed; the bottleneck was not that agents did not know what others needed.

4.2 A liar: public reputation, not private verification

One scripted liar among four cost the honest agents 0.35 welfare per agent-turn. Showing, next to each shared clue, how many of its sender's earlier shares were true (counted over subtasks already solved, so the record is never wrong) recovered 0.32 $[0.20, 0.44]$ of that loss (Holm $p = 0.008$), with no cost when no liar was present (0.006 $[-0.012, 0.026]$). A *check* action that let an agent spend a turn to learn whether a posted clue was true was used 14 times per board and caught 13 false clues, yet recovered nothing (-0.01 $[-0.28, 0.19]$);

a warning instruction did nothing (0.03 $[-0.16, 0.20]$). The check's result stays with the checker and costs a turn; the record reaches everyone for free.

4.3 The hidden-profile vote: aggregation on our vote

The baseline team chose the hidden-best option in 0.54 of classic episodes and 0.88 with a lone dissenter, against a solo ceiling of 1.00. Members posted about half of their private facts (0.50) in every arm. A table of the facts posted so far, grouped by option and without senders, closed 0.75 $[0.50, 1.00]$ of the lone-dissenter gap (win rate 0.97; Holm $p = 0.012$), mainly by changing votes (votes against the best option per episode fell from 0.58 to 0.21), and 0.22 $[-0.02, 0.40]$ in the classic variant. Enforcing facts before opinions in the first round did nothing (-0.02 $[-0.31, 0.18]$). Unlike the board, the prompt-only control was not weaker here: asking members to speak up closed 0.39 $[0.14, 0.59]$ of the classic gap, though not with a lone dissenter.

4.4 Replication on HiddenBench

We reran the vote's arms, unchanged, on HiddenBench's 65 tasks [13]: each task's description, 3 or 4 options, shared facts and one hidden fact per agent, played 3 times with shuffled options and facts (195 episodes per arm), with a plan frozen after a pilot. Our protocol (facts posted verbatim, three simultaneous rounds) differs from HiddenBench's free sequential discussion, and the baseline team was far stronger than the groups the benchmark reports: agent accuracy 0.79 against 30%, with a solo ceiling of 0.91. The fact table did not help (-0.07 $[-0.69, 0.30]$ of the gap), nor did enforcing facts first (0.11 $[-0.37, 0.45]$). The instruction to speak up was the one arm that moved: disclosure of hidden facts rose from 0.74 to 0.88 and the win rate from 0.77 to 0.83 (gap closed 0.41 $[0.00, 0.71]$; a control arm, not Holm-tested). About 3% of replies in every arm were refusals that cut the reply short.

4.5 Which property of trust information matters?

To separate the two ways the track record and the check differ, we added a private record (counted only over subtasks the viewing agent solved itself) and a public check (its result shown to every agent), completing a 2×2 of public or private and free or costly (one liar among four, 10 seeds). The public check recovered 0.14 $[-0.05, 0.30]$ of the liar's damage and the private record 0.13 $[-0.13, 0.29]$, between the public free record and the private costly check. Pooled over the design, publicity added 0.18 $[0.04, 0.36]$ and being free 0.16 $[0.02, 0.30]$; their difference was 0.02 $[-0.16, 0.20]$. Both properties matter, about equally, and they add up. The private record rested on less data than the public one (each agent sees only its own solved subtasks), so for that pair privacy and the amount of information are confounded; the check pair, the same checks shown to one agent or to all, is not.

4.6 Do the fixes scale?

We reran the clearest negative and positive fixes with 8, 16 and 32 agents (5 seeds; a quarter of the agents liars in the liar cells; plans frozen before the runs, the 32-agent cost checked in a pilot). The request board widened the pooling gap by about the same share at every size (-0.16, -0.17, -0.16), and the track record recovered 0.30, 0.22 and 0.31 of the liars' damage; every bootstrap interval excludes zero, though with five seeds the exact sign-flip tests cannot fall below $p = 0.0625$. Larger teams pooled less (pooled share 0.41 at 8 agents, 0.34 at 32) and lost more to the same share of liars (0.47 against 0.54).

Why larger teams lose more to liars. We replayed the baseline runs from the response cache with a turn-by-turn tracer (a further plan frozen before the traces; no API calls) and split the honest agents' loss into additive terms that sum to it exactly. Lies did not become more visible in larger teams, and Haiku agents almost never passed a liar's clue on (at most 0.25% of liar posts). What grew was how often agents built on a lie: the share of their answers that rested on a false clue rose from 0.79 at 8 agents to 0.91 at 32, such answers were wrong 95–97% of the time, and the loss on subtasks a lie touched rose from 0.25 to 0.31 per agent-turn (0.72 of the total rise, interval $[0.16, 3.2]$). Once a liar had posted a clue slot, the true clue was never posted afterwards in 89–94% of cases: agents treat a posted slot as settled, whoever posted it.

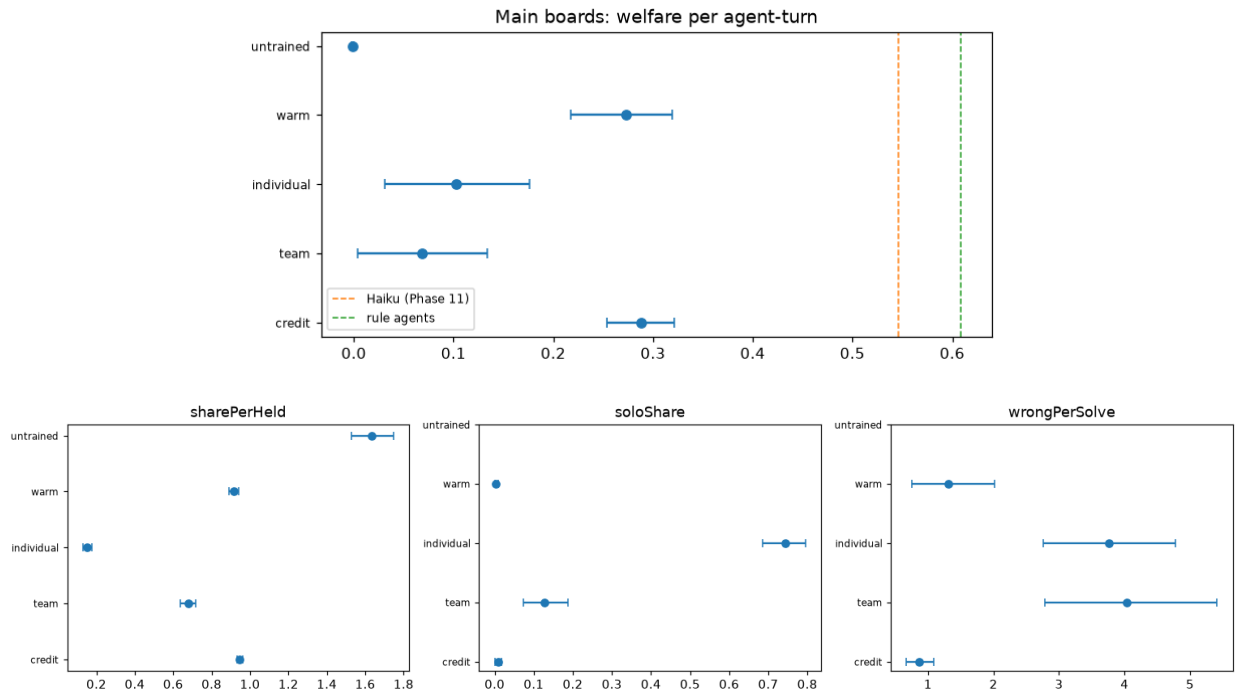


Figure 2: Top: welfare per agent-turn on held-out boards (Haiku and rule agents for reference). Bottom: how models use information: shares per held clue, share of solves by guessing, wrong answers per solve.

5 Incentives when training a team

We trained Qwen2.5-3B-Instruct to play the clue board in teams of four copies sharing one LoRA adapter. A warm start imitated a rule agent's moves; RL then ran 60 steps of group-relative policy gradient with a KL penalty to the warm start (recipes for reasoning RL often drop it [30]) and a learning rate steered to a target per-step policy change, under three rewards: *individual* (an agent's own solves), *team* (every solve shared equally) and *credit* (the solver keeps half; the other half goes to the agents whose shared clues it lacked). Models were evaluated on 15 held-out boards (Figure 2).

Imitation did the teaching: welfare rose from -0.001 to 0.27 $[0.22, 0.32]$. With one training seed per reward, RL added nothing measurable on top: credit 0.02 $[-0.03, 0.06]$, individual -0.17 $[-0.27, -0.07]$, team -0.20 $[-0.26, -0.14]$ relative to the warm start. The rewards differed sharply in what they taught. Under individual reward the model stopped sharing (0.92 to 0.15 shares per held clue) and solved by guessing (74% of solves; 3.8 wrong answers per solve): with a wrong answer costing 0.5, guessing among three candidates for a 3-point subtask earns about 1.25 per turn, more than sharing pays the sharer. Copies of one model with identical weights became selfish because the reward, not the weights, defines their interests. Team reward, with no way to tell whose move helped, was unstable and was halted by our stopping rule. Credit reward kept sharing intact and beat individual by 0.19 $[0.09, 0.27]$ and team by 0.22 $[0.14, 0.29]$; with three trained agents and one Haiku agent it was the best partner (-0.16 against an all-Haiku team; warm start -0.22).

Closing the loophole, and second seeds. A second, pre-registered round trained two seeds each of individual reward with a training-only cost of 3 per wrong answer (`indpen`), credit reward with per-turn advantages [27] and the same cost (`credturn`), and a second seed of credit reward (Figure 3). *Pricing the loophole stopped the selfish drift*, as pre-registered: under the same individual reward, sharing stayed at 1.00 and 0.83 shares per held clue (warm start 0.92) and guessing at 0% and 5% of solves. *RL then beat imitation* with credit: pooled over seeds, credit gained 0.068 $[0.026, 0.112]$, `credturn` 0.069 $[0.018, 0.120]$ and `indpen`, at the edge of significance, 0.053 $[0.000, 0.106]$; the best runs reached 0.39 per agent-turn, 44% above the warm start. *Seeds matter as much as rewards*: the same credit configuration gained 0.02 $[-0.03, 0.06]$ with

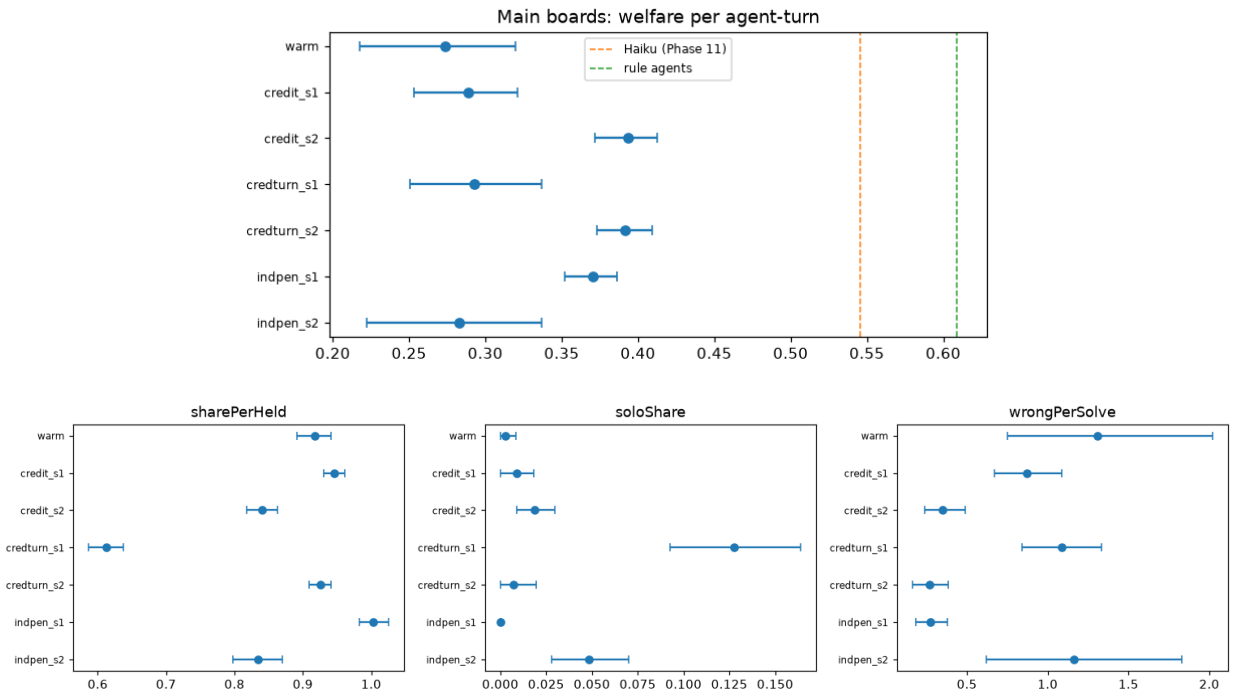


Figure 3: The second round: two training seeds per arm (`credit_s1` is the first round's credit run). Top: welfare per agent-turn on held-out boards. Bottom: information use.

one seed and 0.12 $[0.07, 0.18]$ with another, so the null result above was a one-seed artefact; each arm had one clearly positive seed and one near zero. Per-turn credit added nothing over board-level credit, and its weaker seed drifted toward guessing (13% of solves, 0.61 shares per held clue) despite the cost.

6 Discussion

What worked, and why. Across the three parts, the mechanisms that helped made the team's information easier to trust or to aggregate at no cost in turns (a public track record; a structured view of what was said, though the latter did not replicate on HiddenBench), and the ones that failed added work for each agent (asking, checking, sharing more). This suggests a design principle for agent harnesses: move aggregation and verification into the shared environment rather than into each agent's turn.

Aggregation versus disclosure. Prior diagnoses of hidden-profile failure stress that agents do not surface private information [13]. In our vote, disclosure stayed near one half in every arm, yet a structured table changed the outcome, which suggested weighting what has been said as a separate bottleneck [1, 18]. HiddenBench did not bear this out: there the table did nothing and the arm that helped raised disclosure. The aggregation effect may depend on how the facts are presented (our vote's facts favour or oppose an option explicitly; HiddenBench's must be interpreted), and we no longer claim it in general.

Trust as a public good. A private check produced information worth exactly as much as one agent's next action; a public record produced it for everyone. In our earlier numpy simulations of a liar in a foraging world, private trust weighting was the defence that paid, so the public/private contrast for LLM agents is not a foregone conclusion and deserves direct study.

Incentives and mechanism design. The guessing loophole was a mechanism-design failure rather than only a credit-assignment one: the environment made trial and error cheap. Pricing it kept individually

rewarded agents cooperative, as credit did, so for trained teams the cheapest fix may be in the environment rather than in the reward. The seed-to-seed spread we found (a gain of 0.02 or 0.12 for one configuration) means that conclusions about reward designs from single training runs, including our own first round, should be treated with caution.

Limitations. The tasks are small and synthetic (modular-arithmetic clues; a templated vote that is easier than HiddenBench's); the Claude teams use one model (Haiku 5.5); board arms have 10 seeds (5 in the team-size sweep) and the RL arms one or two training seeds, which our own results show is too few to rank similar designs. Several prompt-only controls were not weaker than affordances in the vote. The vote fixes were retested on HiddenBench [13], where the table did not replicate; the trust fixes have not yet been tested on an established misinformation benchmark [29], and our HiddenBench protocol left little headroom (solo 0.91).

Broader impact. The fixes studied are defensive (against misinformation) or improve cooperation; the RL result shows how easily a plausible incentive teaches agents to abandon cooperation, which matters for anyone training groups of models.

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A Reproducibility

Code, environments and notebooks: https://github.com/Positron00/explRL. Analysis plans were frozen and committed before their runs (Track A of the fixes: `FIX_PLAN`, hash `79abfb531cdd`; RL: `RL_PLAN` v2, hash `4d1ea0078a6e`; Phase 13 RL: `RL13_PLAN`, hash `3e50a4002c7b`; HiddenBench replication: `HB_PLAN`, hash `791a2df079a8`, frozen after a pilot, on the benchmark's 65 tasks vendored unchanged under their MIT licence; trust information: `TRUST_PLAN`, hash `6d2818ae84c2`; team size: `SCALE_PLAN`, hash `b886baefb2ef`, frozen after a pilot; the traces: `TRACE_PLAN`, hash `f5c25637acd0`). All API replies are cached locally (available on request), and cached reruns reproduce every row and turn log byte for byte at no cost. The fixes' main runs made 41,800 Haiku calls in all (including cached baseline cells). Every number in this paper is generated from the saved results by `paper/make_assets.py`.

B Additional observation: a cheap simulation as a screen

Rule-agent simulations in numpy ranked harness designs (channels and team sizes) the way Haiku teams did on the task board (Spearman $\rho = 0.75$ $[0.55, 0.93]$) and the clue board ($\rho = 0.79$ $[0.74, 0.81]$), but not on text foraging ($\rho = 0.02$ $[-0.47, 0.45]$). Surrogates fitted to LLM decisions can predict the macro outcomes of LLM societies [10]; our simulations are not fitted to LLM behaviour, and we study them as a design-screening method in a companion paper. For RL reward designs, simple numpy learners trained under the same five rewards reproduced 7 of 12 effect signs of the first-round runs (all four for individual reward, none for credit reward). Their forecast for the two second-round arms, committed before those runs were evaluated, matched 4 of 8 signs: it predicted that per-turn credit would beat the warm start, but wrongly that pricing the guessing loophole would not stop the selfish drift, an effect a two-decision learner cannot represent.